OFFICIAL ARCHITECTURE SPECIFICATION

Good Bee Skincare Platform

Complete Feature Specification, Role-Based Access & Admin Capabilities Manual

SYSTEM VERSION	INFRASTRUCTURE	CORE ROLES	PORTALS
v2.4 Production Blueprint	Vercel Edge + Render API	Patron & Super Admin	Dealer, Producer, Promoter

1. Executive Architecture Overview

CORE ECOSYSTEM

The **Good Bee Skincare Platform** is an artisanal, cellular skincare commerce engine engineered for luxury brand storytelling and high-frequency operational control. Built on **React 19 + Vite** with an optimized **Node.js / Express** API, it delivers seamless patron journeys and granular administrative command over products, inventory, live CMS, and partner channels.

Patron Storefront (Public) — PUBLIC ACCESS

Ultra-responsive, aesthetic consumer boutique with dual-axis concern & category filters, interactive slide-over cart, GC/MS extraction lab research disclosures, and zero-fee Dynamic UPI QR + COD checkout.

Admin Console (Command Center) — AUTHORIZED

High-frequency floating operational HUD with left-edge mouse-drag resizing, product creation & 1-click formulation cloning, live site CMS broadcasting, order lifecycle dispatch, and financial metrics.

2. Role 1: Patron / Consumer Storefront Capabilities

STOREFRONT EXPERIENCE

The patron experience is designed for botanical transparency, prestige ritual discovery, and frictionless mobile purchasing:

Feature Module	Functionality & Capability	Patron / Business Benefit
Botanical Catalog & Dynamic Filters	Real-time dual-axis filtering by *Concerns* (Youth & Radiance, Barrier Repair, Pigmentation & Tone, Moisture Deficit, Acne) and *Categories* (Pure Essential Oils, Cellular Serums, Pain Roll-Ons, Elixirs).	Sub-second filtering without page reloads; matches patrons to targeted botanical remedies.
Interactive Slide-Over Cart	Persistent slide-over drawer showing quantity incrementors, itemized tally, free shipping eligibility tracker (₹999 tier), and fast checkout launch.	Zero cart abandonment; real-time shipping threshold motivates higher basket size.
Product Detail & Formulation Ritual	Dedicated product dossier displaying GC/MS lab research notes, botanical ingredients, terroir extraction methods, and step-by-step sacred application rituals.	Establishes dermatological/botanical trust, commanding luxury pricing.
Dynamic UPI Intent QR Gateway	Generates an on-the-fly, amount-encoded Dynamic UPI QR (via client/server QRCode engine). Patrons scan with GPay, PhonePe, Paytm, or CRED.	Instant 10-second mobile UPI payment with ₹0 intermediary gateway processing fees.
Cash on Delivery (COD) Option	Traditional payment gateway option with full shipping address verification and automated order tracking integration.	Captures patrons across all regions who prefer cash-on-receipt assurance.
WhatsApp Concierge & Order Dispatch	Pre-formatted WhatsApp order inquiry button pre-populating order ID, patron items, and shipping address to Good Bee's official WhatsApp support channel.	Personalized concierge service and instant human post-purchase support.

3. Role 2: Super Admin Console — Complete Powers & Capabilities (Part I)

FULL CATALOG CONTROL

The **Super Admin Console** is the central nerve center of Good Bee. It provides complete operational autonomy to create, clone, price, modify, and monitor every aspect of the brand without developer assistance.

A. Operational Window Ergonomics & Mouse-Drag Resizability

Mouse-Draggable Left Edge Resizing

CUSTOM HUD

The admin console features a dedicated 6px left-edge grip handle. Admins can click and drag with their mouse to continuously resize the console width from 420px up to 96% of their monitor screen, with a real-time HUD badge display (e.g., `↔ 1040px`).

1-Click Maximize / Restore & Double-Click Expand

ERGONOMICS

A dedicated `Maximize2` / `Minimize2` icon toggles between comfortable side-docked review and true fullscreen command view. Double-clicking the resize handle toggles between maximum and default width instantly.

B. Complete Product Catalog & Formulation Controls

Admins possess full CRUD (Create, Read, Update, Delete) and rapid prototyping powers over the product catalog:

Admin Capability	Action & Scope of Control	Trigger / Location
Add New Botanical Product	Full formulation wizard: Title, Subtitle, Slug, SKU, Category, Selling Price (₹), MRP (₹), Volume/Weight, Initial Stock, Low-Stock Threshold, Marketing Badge, Short & Long Descriptions, Ingredients list, Sacred Ritual, Research Notes, and Primary/Gallery images.	`+ Add New Formulation` button in Products tab.
1-Click Product Cloning / Duplication	Instantly creates a new draft formulation by cloning all specifications, ingredients, ritual steps, and gallery links of an existing product, appending `-copy` to SKU and slug for fast SKU variant rollout.	`Copy` icon button on each product card.
Dynamic Custom Categories	Admin can select from standard categories or pick *"+ New Category"* to type any custom category name on the fly. The new category instantly reflects across the storefront filters.	Category selector in Product Editor.
Real-Time Pricing & Stock Updates	Adjust active selling price, original MRP (for discount percentage displays), and physical bottle count in inventory with immediate storefront synchronization.	Product Edit Modal & Quick Stock adjuster.
Configurable Low-Stock Thresholds	Define custom minimum stock levels per SKU (e.g. alert when below 10 units). Automatically calculates inventory health and flags warnings on the Admin dashboard.	`lowStockThreshold` field in product modal.
Marketing Badge Tagging	Assign luxury storefront badges: *'100% Wild Harvested'*, *'Bestseller'*, *'Clinical Formulation'*, *'Limited Batch'*, or custom labels.	Badge selector in Product Editor.
Formulation Ritual & Lab Notes Editor	Live textarea controls to author the exact patron skincare ritual and publish technical GC/MS distillation analysis notes directly to the product page.	Ritual & Research Notes tabs.
Product Deletion with Safeguard	Remove discontinued or seasonal items from the live storefront with confirmation dialog protection to prevent accidental loss.	`Trash2` icon on product card.

3. Role 2: Super Admin Console — Complete Powers & Capabilities (Part II)

CMS & COMMERCIAL CONTROL

C. Live Site Content Management System (CMS)

The admin can update site copy, branding, ethics narratives, and reviews with zero code redeployments:

CMS Control Module	Editable Content & Live Broadcast Effect	Admin Tab
Announcement Banner	Change top bar ticker text (e.g., *"Complimentary Wild-Harvest Frankincense with orders above ₹2,500"*) and toggle visibility on/off across the site.	`Site Content → Announcement`
Hero Title & Narrative	Live rewrite of the storefront main header title, subtitle, and primary call-to-action button text without code redeployment.	`Site Content → Hero`
Brand Philosophy Statement	Edit the brand's core bio-botanical cellular manifesto and sacred botanical harvesting narrative in the "Our Philosophy" section.	`Site Content → Philosophy`
Sourcing & Terroir Lab Narrative	Update sustainability credentials, steam-distillation purity promises, and wildcrafting ethical sourcing disclosures.	`Site Content → Sourcing`
Customer Reviews & Testimonials	Add, edit, or remove customer reviews: patron name, star rating (1–5), review quote, and botanical formulation verified.	`Site Content → Reviews`

D. Promotions, Discount Coupons & Payment Gateway Orchestration

Coupons & Promotion Engine

DISCOUNTS

- **Create Coupons:** Set custom coupon code (e.g., `ROYALBEE20`).
- **Discount Schemes:** Choose Percentage (e.g., 20% off) or Flat Rupee amount (e.g., ₹300 off).
- **Cart Rules:** Enforce minimum cart spend thresholds and custom expiration dates.
- **Status Controls:** Instant toggle between Active and Inactive states, or permanent deletion.

Payment Gateway Orchestrator

PAYMENT CONTROL

- **Dynamic UPI QR Toggle:** Enable or disable instant UPI payments site-wide with 1 click.
- **VPA / Merchant Configuration:** Update target UPI VPA address (e.g., `goodbee.official@okaxis`) and merchant payee display name.
- **Cash on Delivery (COD) Switch:** Enable or disable COD checkout instantly based on delivery logistics coverage.

3. Role 2: Super Admin Console — Complete Powers & Capabilities (Part III)

OPERATIONS & PORTALS

E. Real-Time Financial Analytics & Order Fulfillment Engine

Admins monitor real-time business health and manage customer shipments directly through the console:

Dashboard KPI	Calculated Metric & Pipeline	Operational Value
Gross Revenue (GMV)	Live tally of all confirmed customer orders in INR (₹) across UPI and COD channels.	Instant macro financial pulse.
Order Volume Tracker	Breakdown of *Total Orders*, *Completed*, and *Pending Verification*.	Detects fulfillment bottlenecks.
Active Patron Database	Unique customer account and order profile counts with repeat patron metrics.	Monitors customer retention & growth.
Commission Liabilities	Accumulated affiliate commission tracking owed to registered brand promoters.	Maintains partner payout accuracy.

ORDER LIFECYCLE MANAGEMENT WORKFLOW

From the **Orders Tab**, the admin can inspect the complete shipping dossier for every patron (Full Name, Phone Number, Shipping Address, City, Postal Code, Itemized Order Lines, and Payment Method). Admins advance orders through the complete lifecycle:

PENDING VERIFICATION → PROCESSING & PACKING → DISPATCHED → DELIVERED (or CANCELLED).

1-Click WhatsApp Dispatch: Admin can click the WhatsApp icon next to any order to trigger an automated dispatch notification to the patron's mobile number with tracking info.

4. Partner Portals (Dealer, Producer, Promoter)

ECOSYSTEM ROLES

In addition to the Super Admin, the Good Bee platform features three dedicated partner portal roles:

Dealer Portal (B2B Wholesale) WHOLESALE

Allows luxury spas, dermatological clinics, and authorized retailers to place bulk orders, access wholesale discount tiers (30%–45% off retail), view minimum order quantities (MOQ), and request GST invoices.

Producer Portal (Extraction Labs) HARVEST TRACE

Empowers wild-harvesting beekeepers and hydro-distillation laboratories to register harvest batches, upload GC/MS purity test certificates, log distillation temperatures, and generate authenticity batch QR seals.

Promoter Portal (Affiliate & Brand Ambassador Network) AFFILIATES

Equips skincare influencers, holistic wellness ambassadors, and beauty practitioners with personalized tracking codes (e.g., `GB-LAVENDER10`), live conversion dashboards, payout balances, and real-time commission statements.

5. Complete Role Permissions & Capability Matrix

SECURITY & ACCESS CONTROL

Comprehensive matrix demonstrating the clear segregation of duties and access permissions across all platform roles:

Capability / Feature Module	Patron	Super Admin	Dealer (B2B)	Producer (Lab)	Promoter
Browse Catalog & Botanical Filters	✓ Full	✓ Full	✓ Wholesale	✓ Read	✓ Read
Add to Cart & Consumer Checkout (UPI / COD)	✓ Full	✓ Full	—	—	—
Draggable & Resizable HUD Portal Window	—	✓ Full	✓ Modal	✓ Modal	✓ Modal
Add Products & Edit Any Specification	—	✓ Full	—	—	—
1-Click Product Formulation Cloning	—	✓ Full	—	—	—
Live CMS: Announcement, Hero, Sourcing, Reviews	—	✓ Full	—	—	—
Manage Discount Coupons & Promotions	Apply Only	✓ Full	—	—	—
Payment Gateways Toggle & UPI VPA Customizer	—	✓ Full	—	—	—
View Financial Revenue & GMV Metrics	—	✓ Full	—	—	—
Order Status Lifecycle & WhatsApp Dispatch	View Own	✓ Full	View Own B2B	—	—
Inventory Health & Low-Stock Alerts	—	✓ Full	Stock Level	—	—
Submit Batch Extraction & Purity Records	—	✓ Review	—	✓ Full	—
Track Referral Links & Affiliate Commissions	—	✓ Full	—	—	✓ Own Stats

6. Architecture & Scalability Blueprint

DEVOPS & CLOUD

Frontend Layer: Vercel Edge

ZERO-LATENCY

Single Page Application served from Vercel's global CDN edges with sub-40ms response times and instant zero-downtime cache invalidation.

Backend Layer: Render Microservice

REST API

Node.js Express API service configured via `render.yaml` blueprint. Supports JWT auth, order routing, and modular database scalability.

Resilient Dual-Sync Engine (Cold Start & Offline Immunity)

PATENTED RELIABILITY

Integrated via `mockBackend.js`: When deployed to production, the storefront automatically prioritizes the live Render API. If the backend is cold-starting or temporarily offline, the frontend seamlessly fails over to client-side localStorage state, guaranteeing that no customer ever encounters a blank screen or broken checkout.